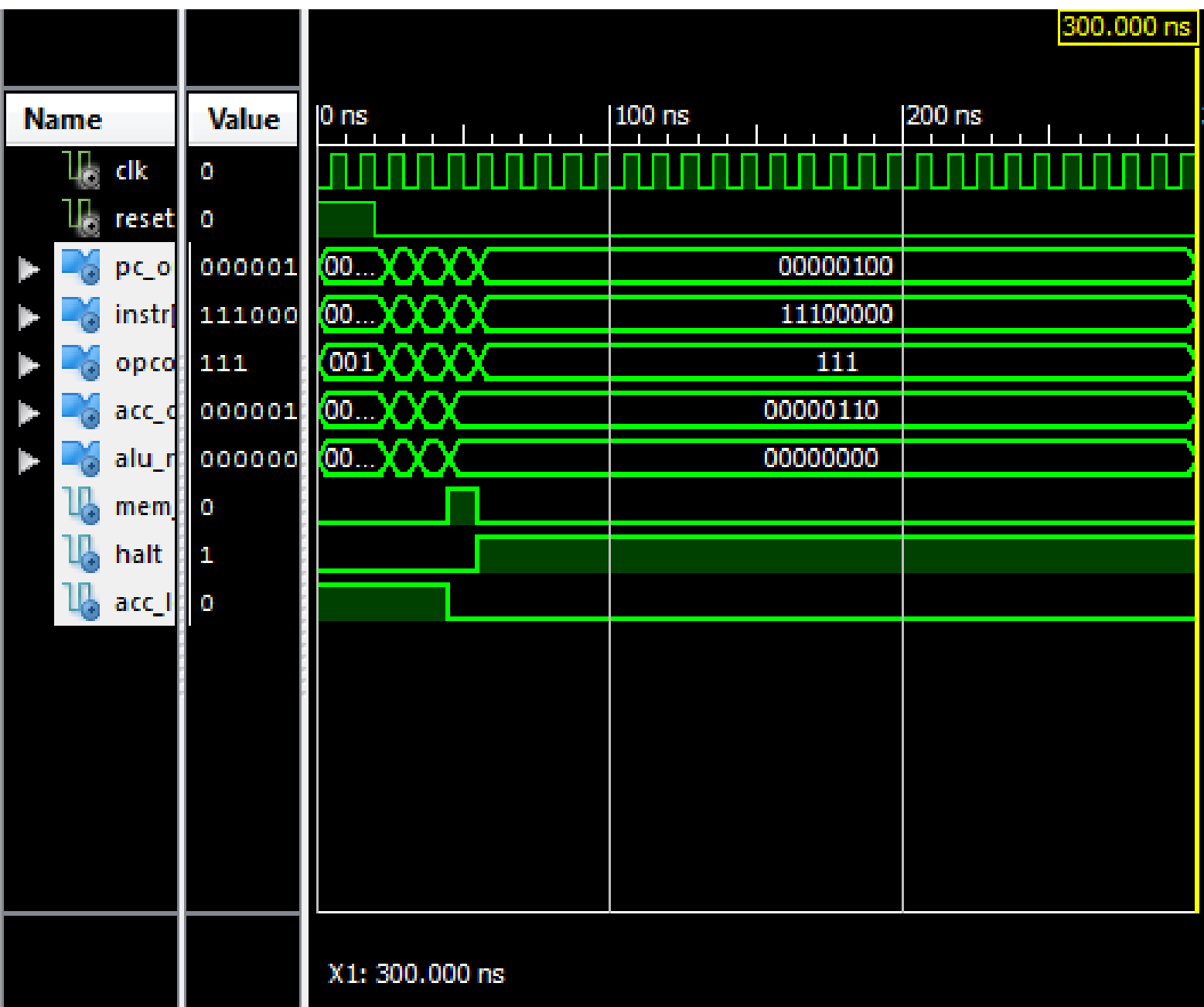


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CPU Design Using VHDL

Figure 1: CPU Execution Waveform



The waveform shows the CPU executing instructions and updating the accumulator until the HALT instruction stops execution.

Figure 2: RAM Memory Contents

Objects		
Simulation Objects for U_RAM		
I O I/O + C U		
Object Name	Value	Data Type
I clk	0	Logic
I we	0	Logic
I ▶ addr[4:0]	00000	Array
I ▶ data_in[7:0]	00000110	Array
O ▶ data_out[7:0]	00000000	Array
+ ▶ memory[0:31]	00000000, 000001...	Array

The RAM memory window shows the value stored after execution of the STORE instruction.